Instituția: Universitatea Tehnică Din Cluj Napoca
Facultatea: Facultatea de Electronică, Telecomunicații și Tehnologia Informației
Disciplina: Sisteme cu Circuite Integrate Analogice (SCIA)

PROIECT SCIA

Student: ANDRONE HORAȚIU GLIGOR

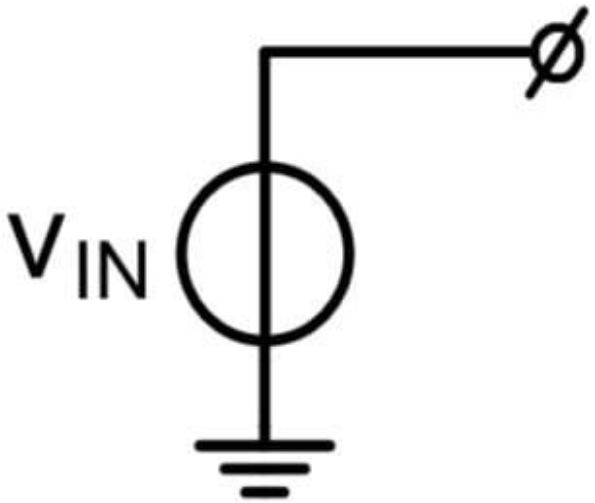
TEMATICA PROIECTULUI

Proiect are ca scop realizarea și analiza unui sistem electronic bazat pe sisteme cu circuite integrate analogice (SCIA), destinat prelucrării unui semnal electric analogic, utilizând amplificatoare operaționale și circuite electronice.

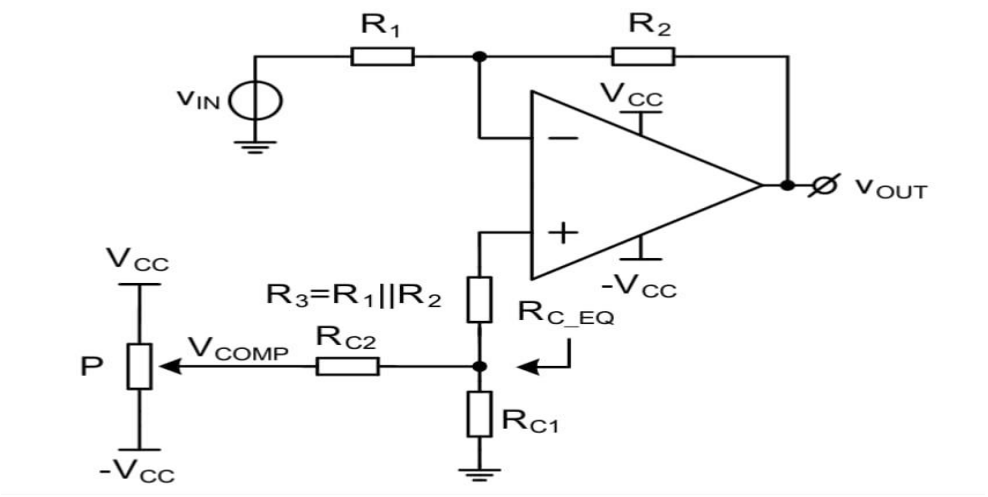
Sistemul este structurat în patru etaje funcționale, realizate cu amplificatoare operaționale, fiecare având parametri electrici specifici. Proiectul analizează etapele de prelucrare a semnalului analogic, incluzând condiționarea, filtrarea activă, amplificarea cu câștig selectabil și redresarea de precizie, evidențiind rolul circuitelor integrate analogice în aplicațiile de prelucrare a semnalelor.

Specificații generale

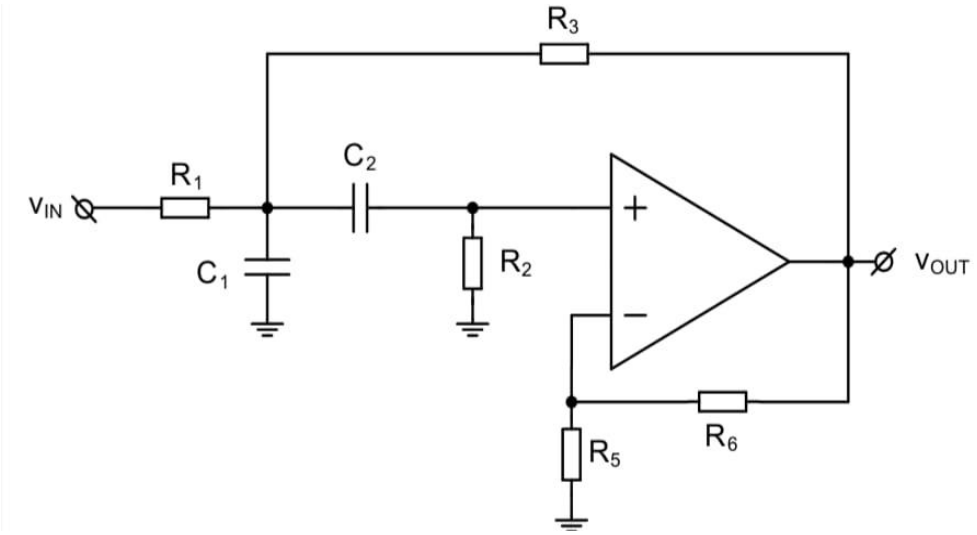
- Sursa de semnal – single ended
  - Amplitudine minimă: 199mV
  - Amplitudine maximă: 561mV



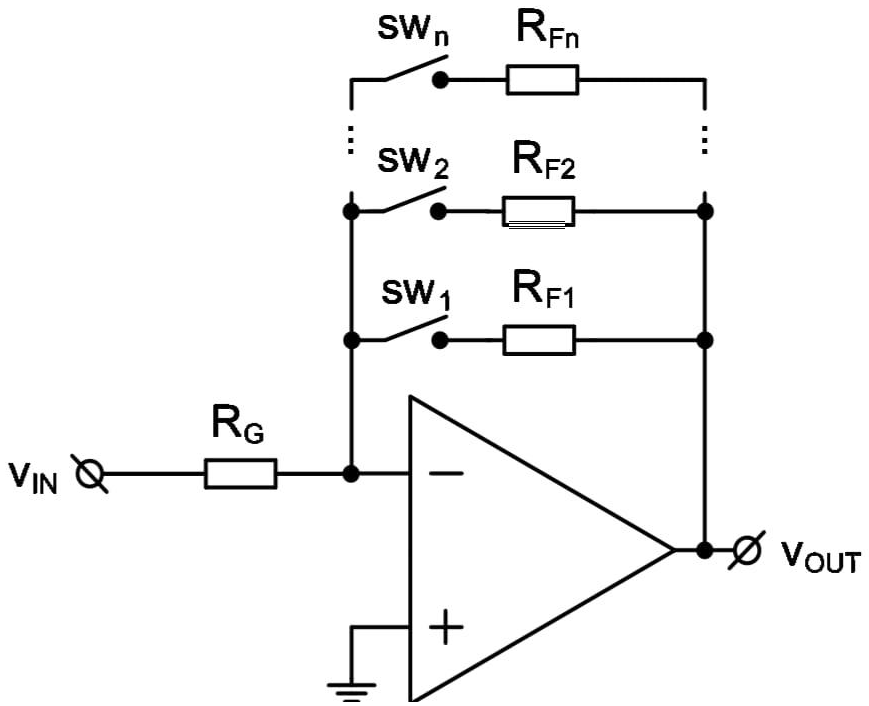
- Etaj 1: AO + compensare DC V-V – Amplificator inversor
  - Câștig liniar: 7
  - Rin_min: 1 kOhm
  - Reglaj Out_DC: DA



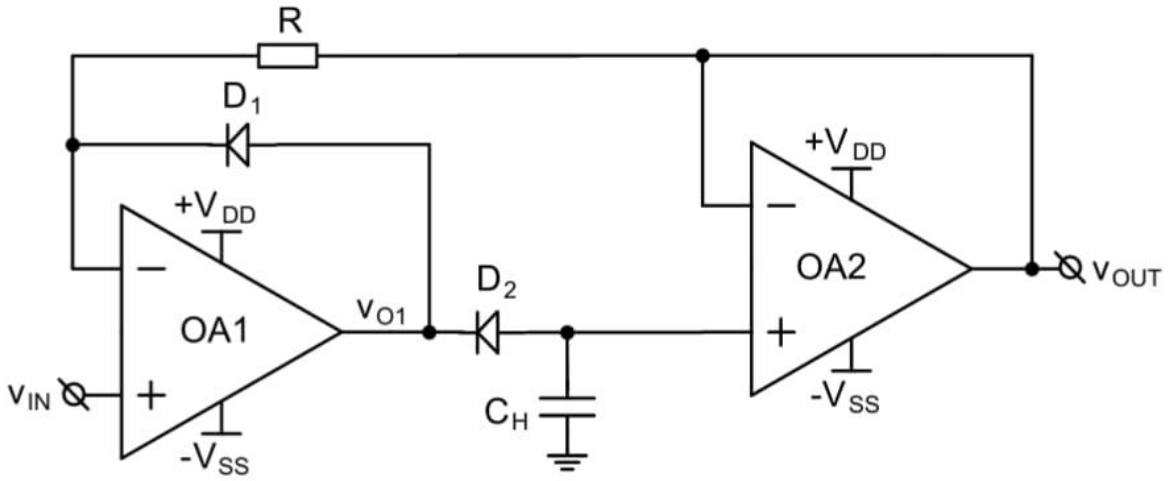
- Etaj 2: Filtru Trece-Banda Sallen-Key
  - Câștig liniar: 1
  - Rin_min: 750 ohm
  - Banda: 1 kHz
  - Q: 0,707



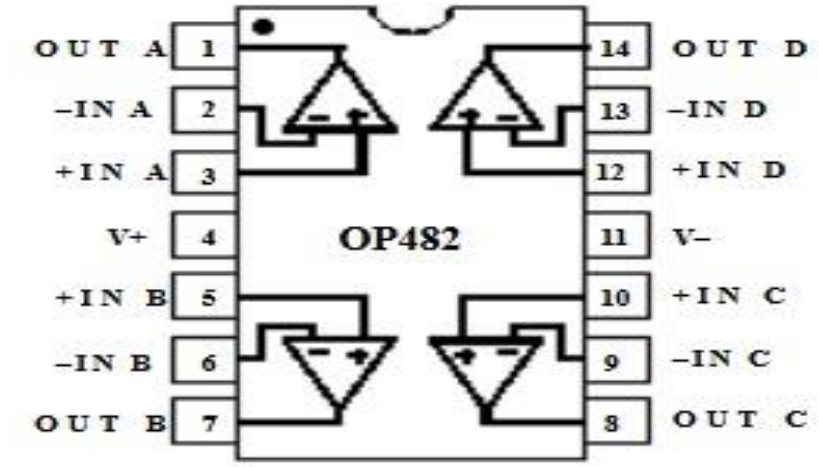
- Etaj 3: Amplificator cu câștig reglabil prin comutatoare
  - Câștig minim: -1 dB
  - Rezoluție: 3dB
  - Număr pași: 4
  - Câștig maxim: 8 dB
  - Rin_min: 3 kOhm



- Etaj 4: Detector de vârf negativ
  - Câștig liniar: 1



- Amplificator operational: OP482



https://www.alldatasheet.com/html-pdf/49049/AD/OP482/43/2/OP482.html

Dimensionarea etajelor

- Etaj 1 – dimensionare:

$Av = -\frac{R2}{R1}$ = 7

$$R1 = 10k\Omega$$

$$R_2 = |A_v| \cdot R_1 = 7 \cdot 10k\Omega = 70k\Omega$$

$$R_3 = R_1 \parallel R_2 = \frac{R_1 \cdot R_2}{R_1 + R_2} = 8{,}75k\Omega$$

$$V_+ = V_{COMP} \cdot \frac{RC1}{RC1 + RC2}$$

$$Vout = -\frac{R_2}{R_1} \cdot Vin + \left(1 + \frac{R_2}{R_1}\right) \cdot V +$$

$$\Rightarrow RC1 = 1k\Omega \qquad RC2 = 220k\Omega$$

- Etaj 2 – dimensionare:

Conform:

f0 [Hz]	w0[rad/sec]	Q	H0[V/V]	H0[dB]	k
1.00E+03	6.28E+03	7.07E-01	1.00E+00	-2.62E-03	2.00E+00

set C1=C2=C [F]	set R1= R2=R3=R[ohm]	set RA [ohm]	RB
1.00E-07	3.18E+03	6.80E+03	6.80E+03

$$H_{BP} = \frac{Vout_{BP}}{Vin} = \frac{(s \cdot wz)}{s^2 + s \cdot \frac{wp}{Q} \cdot wp^2}$$

$$R1 = R2 = R3 = 3{,}18k\Omega$$

$$RA = RB = 6{,}8k\Omega$$

- Etaj 3 – dimensionare:

SW 1	SW 2	SW3	SW 4	RG	RF	AV_liniar
1	0	0	0	RG	R4	0,891
0	1	0	0	RG	R3	1,259
0	0	1	0	RG	R2	1,778
0	0	0	1	RG	R1	2,512

$$AdB = [-1,\ 2,\ 5,\ 8]$$

$$AV_{liniar} = 10\ \frac{AdB}{20}$$

$$AV = \frac{RG}{RF} \Rightarrow RF = AV \cdot RG$$

$$Aleg\ RG = 3{,}3k\Omega$$

$RF1 = 2{,}94k\Omega \qquad RF2 = 4{,}15k\Omega \qquad RF3 = 5{,}67k\Omega \qquad RF4 = 8{,}29k\Omega$

- Etaj 4 – dimensionare:

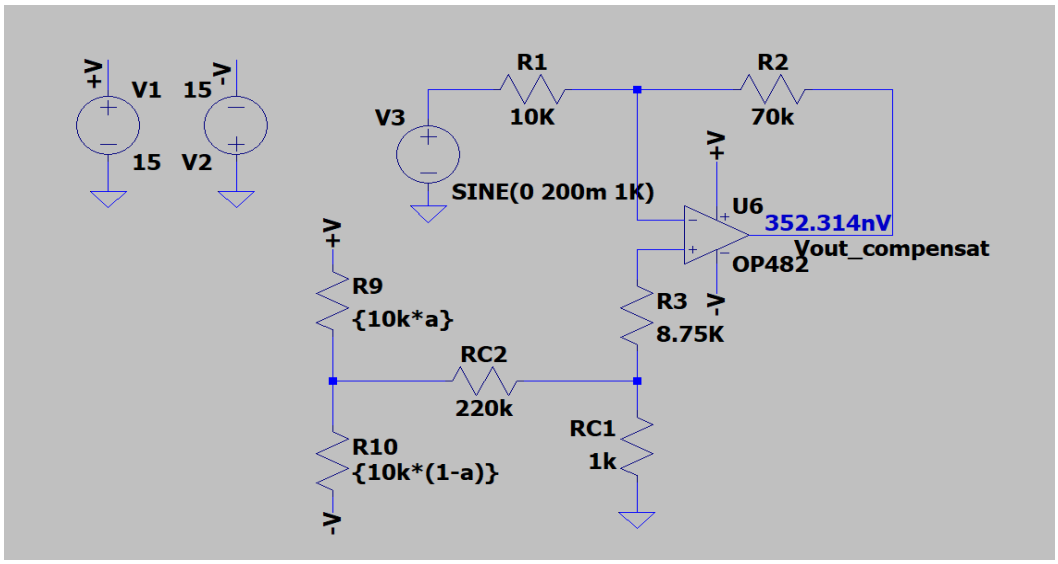
$Vin > 0 \Rightarrow VOA1 \uparrow VDD \Rightarrow D1\ ON,\ D2\ OFF \Rightarrow VO = -\frac{R2}{R1} \cdot Vin$

$Vin < 0 \Rightarrow VOA2 \downarrow -VSS \Rightarrow D1\ OFF,\ D2\ ON \Rightarrow VO = Vin$

$R = 10k\Omega \qquad C = 100nF$

Caracterizare etajelor

- Etaj 1:



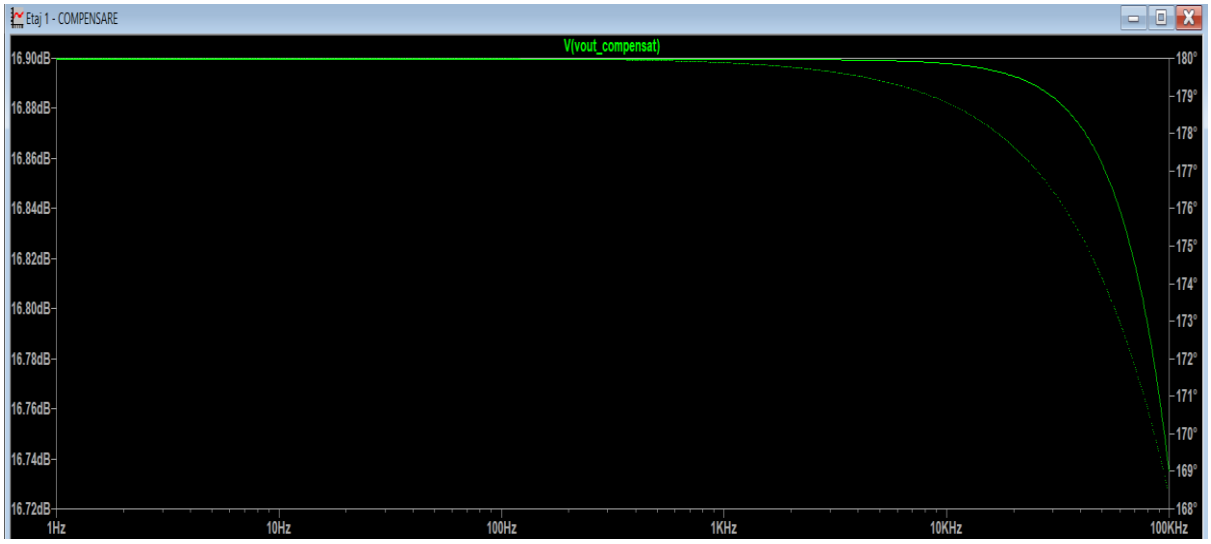
$$a = 501.73m$$

-Simulare DCOP:

C:\Users\Lenovo\Desktop\.Proiect - SIMULARI\Etaj 1 - COMPENSARE.net

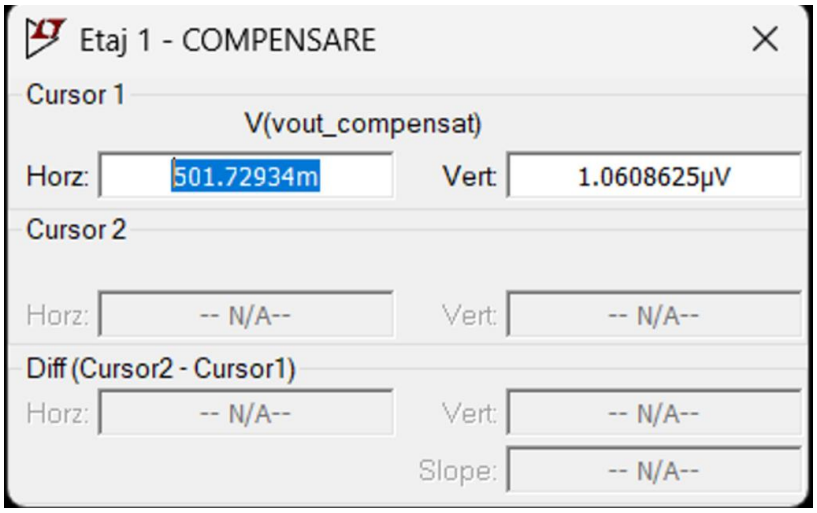
```
       --- Operating Point ---

V(-v):          -15             voltage
V(n001):        0               voltage
V(n003):        -0.000232197    voltage
V(n004):        -0.0513195      voltage
V(+v):          15              voltage
V(n002):        5.01592e-08     voltage
V(n005):        -0.000232213    voltage
V(vout_compensat):              3.52314e-07    voltage
I(V1):          -0.00400401     device_current
I(RC2):         -2.32215e-07    device_current
I(R10):         0.00300012      device_current
I(R2):          4.3165e-12      device_current
I(R1):          5.01592e-12     device_current
I(RC1):         2.32213e-07     device_current
I(R3):          1.82048e-12     device_current
I(R9):          0.00299988      device_current
I(V2):          -0.00400425     device_current
I(V3):          5.01592e-12     device_current
Ix(u6:1):       -1.78893e-12    subckt_current
Ix(u6:2):       -6.99418e-13    subckt_current
Ix(u6:3):       0.00100413      subckt_current
Ix(u6:4):       -0.00100413     subckt_current
Ix(u6:5):       -4.3165e-12     subckt_current
```

-Simulare: Câștig la joasă frecvență

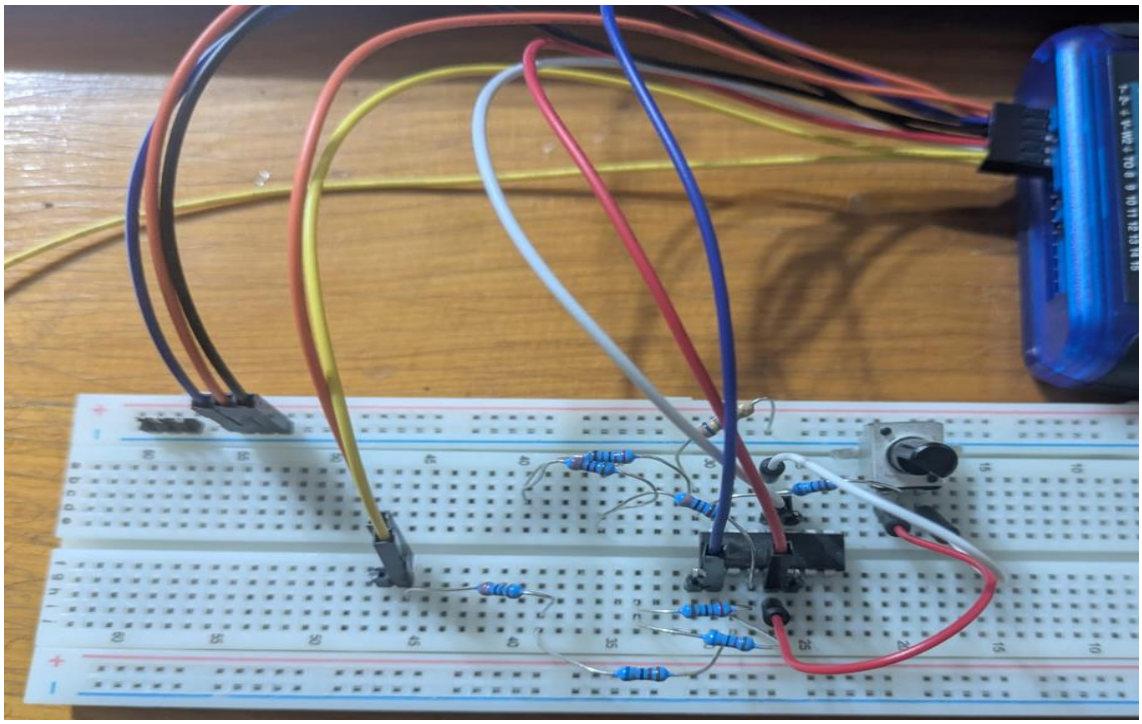


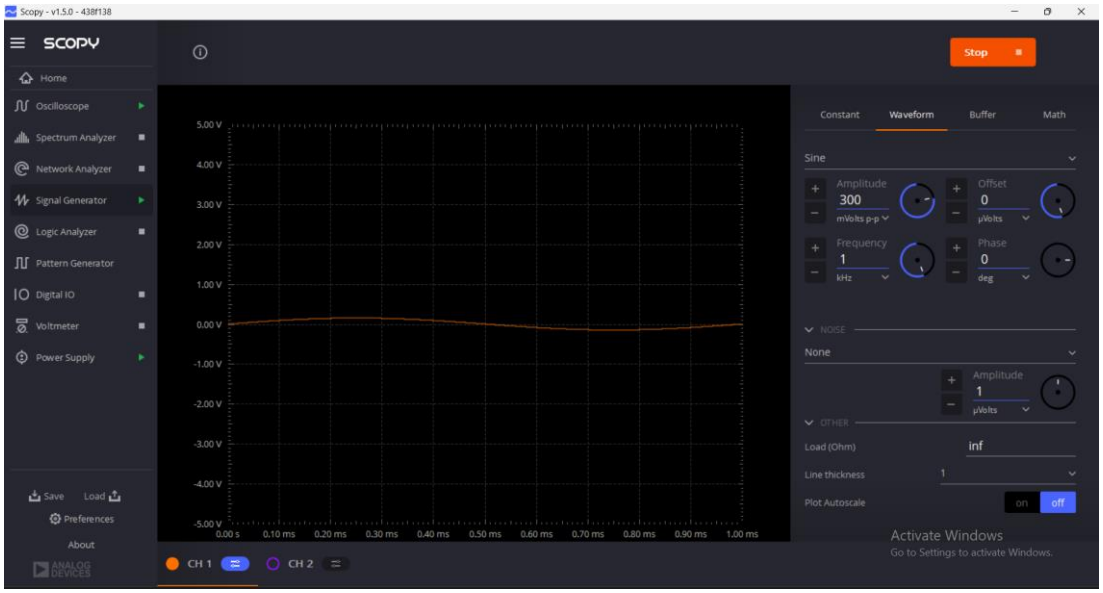
$$Av_{liniar} = 10\,\frac{AdB}{20} = 10\,\frac{16.9}{20} \sim 7$$

-Simulare: Compensarea



Simulare Circuit Real:





Scopy - v1.5.0 - 438f138
SCOPY
Print
Enable Mixed Signal View
Stop
Single
Home
Oscilloscope
Spectrum Analyzer
Network Analyzer
Signal Generator
Logic Analyzer
Pattern Generator
Digital IO
Voltmeter
Power Supply
Save
Load
Preferences
About
500.000 µs/div
8 k Samples at 1 Msps
Triggered
500 mV/div (±2.5)
500 mV/div (±2.5)
CH 1
CH 2
Cursors
Measure
Trigger
Activate Windows
Go to Settings to activate Windows.

Scopy - v1.5.0 - 438f138
SCOPY
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Enable Mixed Signal View
Stop
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Home
Oscilloscope
Spectrum Analyzer
Network Analyzer
Signal Generator
Logic Analyzer
Pattern Generator
Digital IO
Voltmeter
Power Supply
Save
Load
Preferences
About
500.000 µs/div
8 k Samples at 1 Msps
Triggered
-3.874 ms,2.069 V
500.000 mV/div (±2.5)
500.000 mV/div (±2.5)
CH 1
CH 2
Cursors
Measure
Trigger
Channel 1
HORIZONTAL
Time Base
500
µs
Position
0
ns
VERTICAL
Volts/Div
500
mVolts
Position
0
µVolts
SETTINGS
CH Thickness
Curve Style
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Lines
Memory depth
Probe Attenuation
Activate Windows
Go to Settings to activate Windows.

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SCOPY
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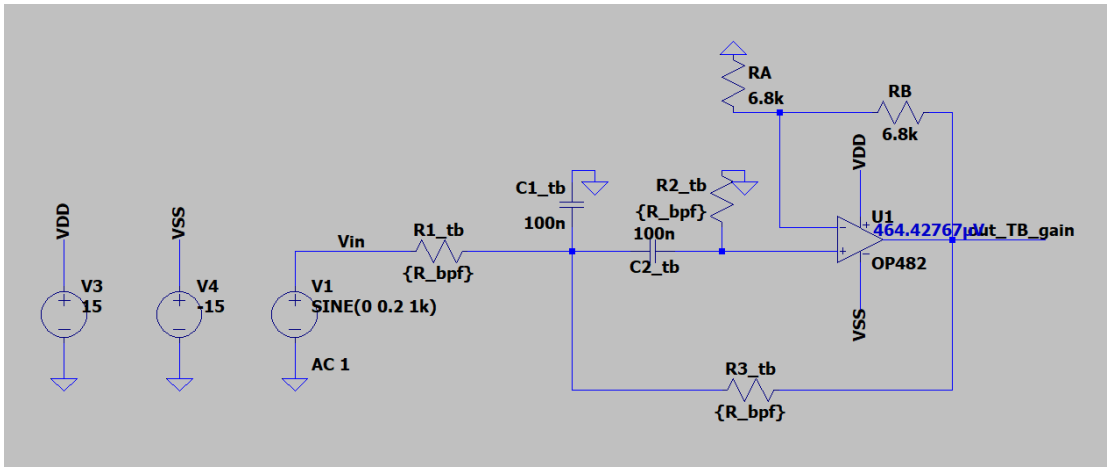
Spectrum Analyzer
Network Analyzer
Signal Generator
Logic Analyzer
Pattern Generator
Digital IO
Voltmeter
Power Supply
Save
Load
Preferences
About
500.000 µs/div
8 k Samples at 1 Msps
Triggered
500.000 mV/div (±2.5)
500.000 mV/div (±2.5)
CH 1
CH 2
Cursors
Measure
Trigger
Channel 1
HORIZONTAL
Time Base
500
µs
Position
0
ns
VERTICAL
Volts/Div
500
mVolts
Position
0
µVolts
SETTINGS
CH Thickness
Curve Style
1
Lines
Memory depth
Probe Attenuation
Activate Windows
Go to Settings to activate Windows.

- Etaj 2:

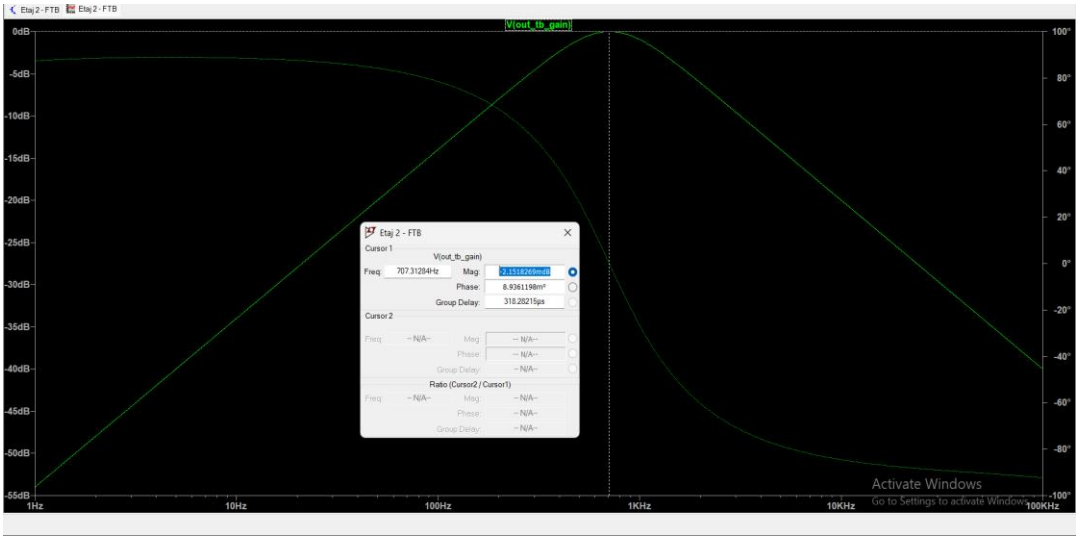


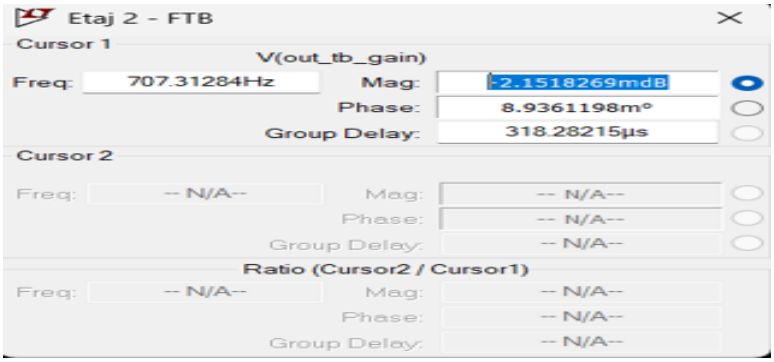
Simulare: DCOP

C:\Users\Lenovo\Desktop\.Proiect - SIMULARI\Etaj 2 - FTB.net

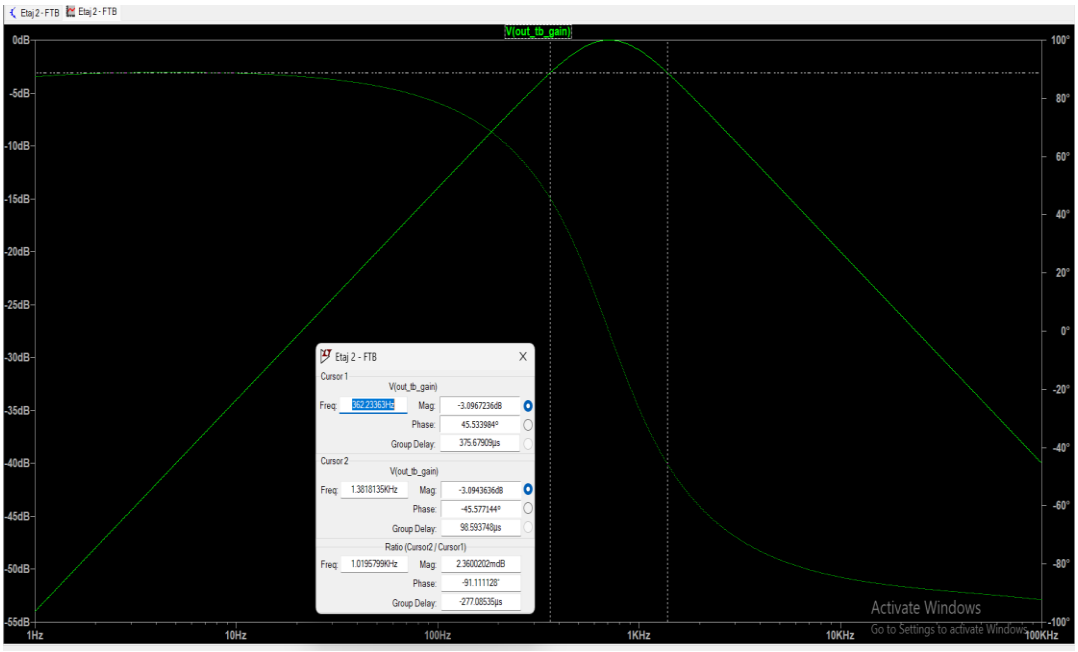
```
       --- Operating Point ---

V(vss):          -15            voltage
V(n002):         0.000232214    voltage
V(n003):         -2.5129e-09    voltage
V(n001):         0.000232208    voltage
V(vdd):          15             voltage
V(vin):          0              voltage
V(out_tb_gain):                 0.000464428    voltage
I(V1):           7.30232e-08    device_current
I(R2_tb):        -7.90222e-13   device_current
I(R3_tb):        7.30232e-08    device_current
I(V3):           -0.00100413    device_current
I(R1_tb):        7.30232e-08    device_current
I(C1_tb):        -2.32214e-23   device_current
I(RB):           3.415e-08      device_current
I(V4):           0.00100413     device_current
I(C2_tb):        2.32216e-23    device_current
I(RA):           -3.41482e-08   device_current
Ix(u1:1):        7.90222e-13    subckt_current
Ix(u1:2):        1.84314e-12    subckt_current
Ix(u1:3):        0.00100413     subckt_current
Ix(u1:4):        -0.00100413    subckt_current
Ix(u1:5):        -1.07173e-07   subckt_current
```

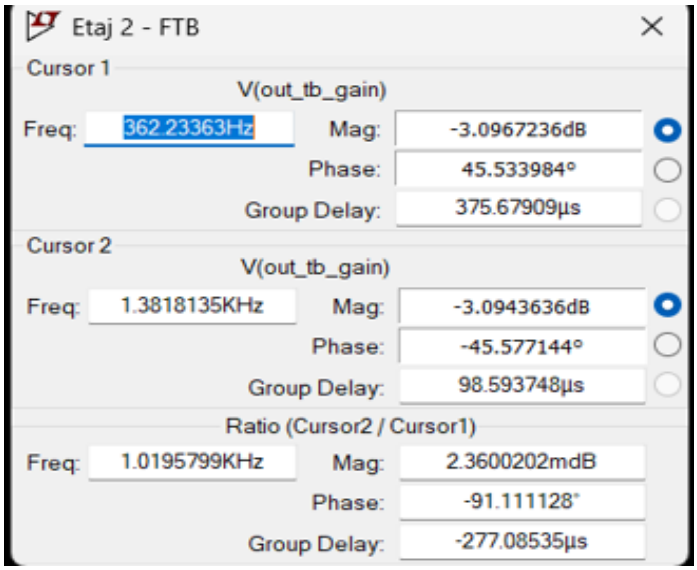
Simulare: Câștig





Simulare: Banda



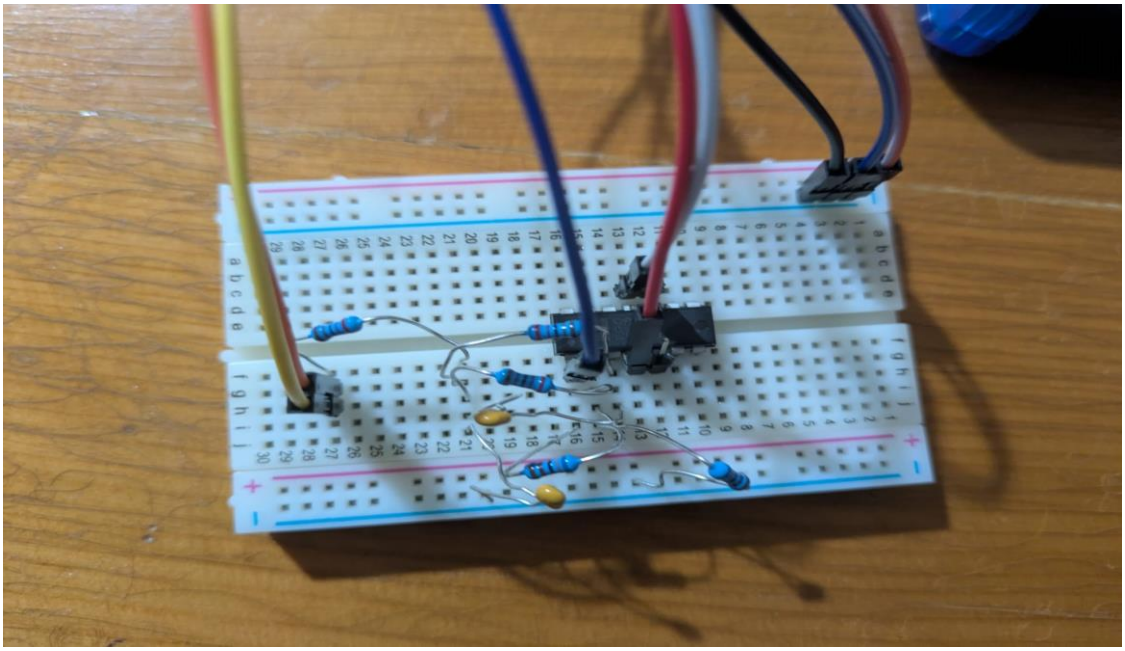


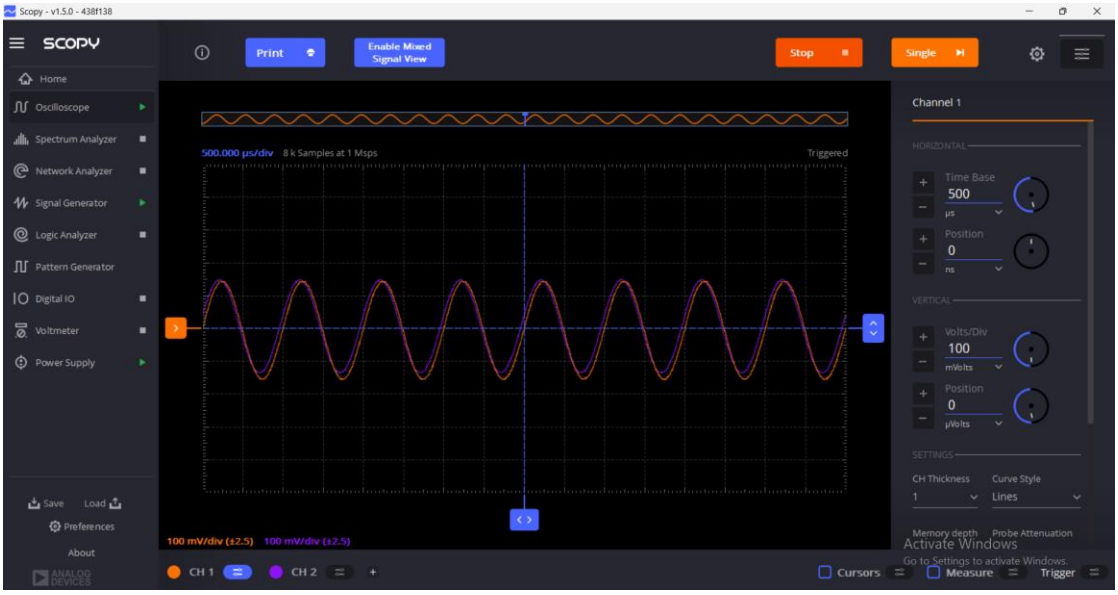
Simulare: Liniaritatea

SPICE Output Log: C:\Users\Lenovo\Desktop\.Proiect - SIMULARI\Etaj 2 - FTB.log

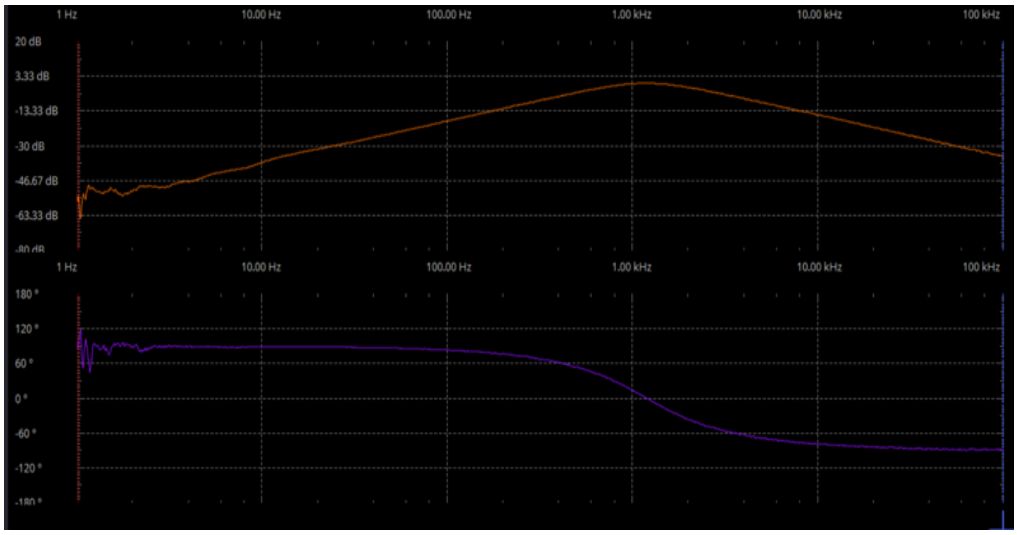
```
LTspice 24.1.8 for Windows
Circuit: C:\Users\Lenovo\Desktop\.Proiect - SIMULARI\Etaj 2 - FTB.net
Start Time: Fri Jan  9 18:48:00 2026
Options: plotwinsize=0  numdgt=7
solver = Normal
Maximum thread count: 16
tnom = 27
temp = 27
method = modified trap
Direct Newton iteration for .op point succeeded.
Total elapsed time: 0.365 seconds.

Files loaded:
C:\Users\Lenovo\Desktop\.Proiect - SIMULARI\Etaj 2 - FTB.net
C:\Users\Lenovo\AppData\Local\LTspice\lib\sub\OP482.lib

Fourier components of V(out_TB_gain)
N-Period=1
DC component:0.000462495

Harmonic          Frequency          Fourier          Normalized          Phase
 Number             [Hz]            Component          Component         [degree]
    1              1.000e+03         1.789e-01          1.000e+00          116.5
    2              2.000e+03         3.859e-06          2.158e-05           32.0
    3              3.000e+03         4.844e-05          2.709e-04          127.0
    4              4.000e+03         3.330e-06          1.862e-05         -106.9
    5              5.000e+03         4.036e-05          2.257e-04          -28.5
    6              6.000e+03         2.524e-06          1.411e-05          117.9
    7              7.000e+03         3.222e-05          1.802e-04          176.9
    8              8.000e+03         2.122e-06          1.186e-05           -8.9
    9              9.000e+03         2.498e-05          1.397e-04           22.9
   10              1.000e+04         2.138e-06          1.196e-05         -128.4
Partial Harmonic Distortion: 0.042135%
Total Harmonic Distortion:   0.000000%
```

Simulare: Circuit Real

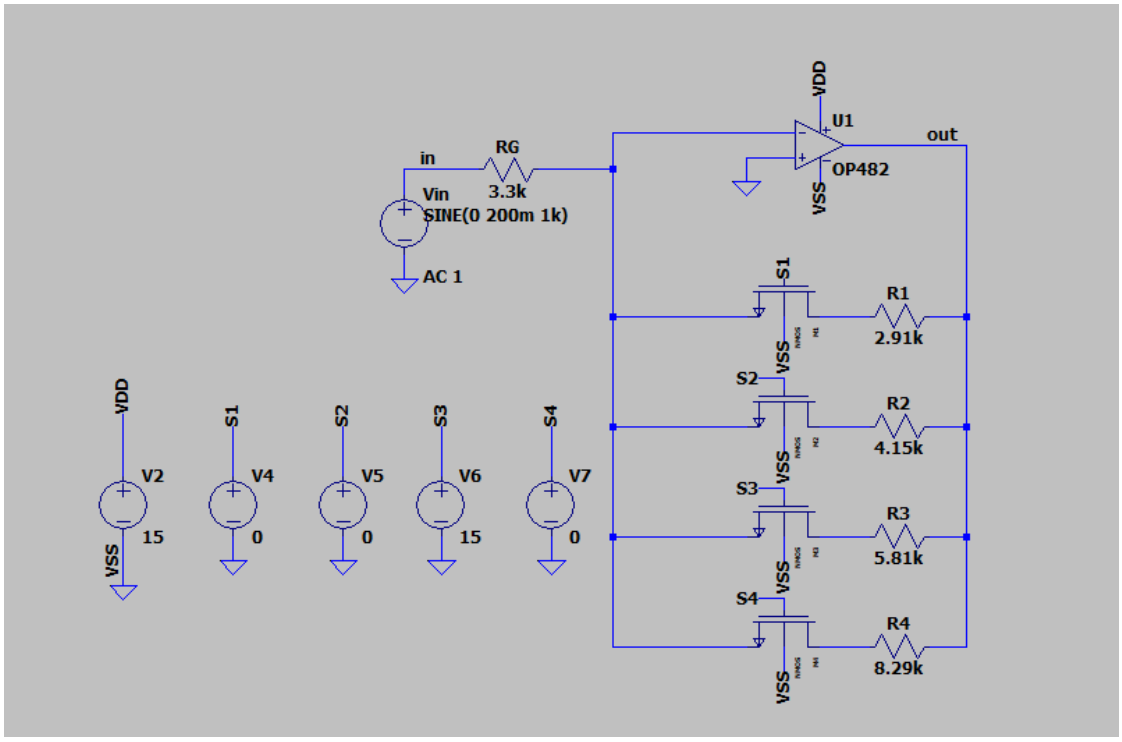




Scopy - v1.5.0 - 438f138
SCOPY
Print
Stop
Single
Home
Oscilloscope
Spectrum Analyzer
Network Analyzer
Signal Generator
Logic Analyzer
Pattern Generator
Digital IO
Voltmeter
Power Supply
Save
Load
Preferences
About
Sample: 1000 / 1000
Current Frequency: 10.000 kHz
Average: 1 / 1
DC Voltage: 4.76 mV
Gain Mode: High
Running
59.1650 Hz, 7.44 dB
Activate Windows
Cursors


1 Hz
10.00 Hz
100.00 Hz
1.00 kHz
10.00 kHz
100 kHz
20 dB
3.33 dB
-13.33 dB
-30 dB
-46.67 dB
-63.33 dB
180 °
120 °
60 °
0 °
-60 °
-120 °

- Etaj 3:

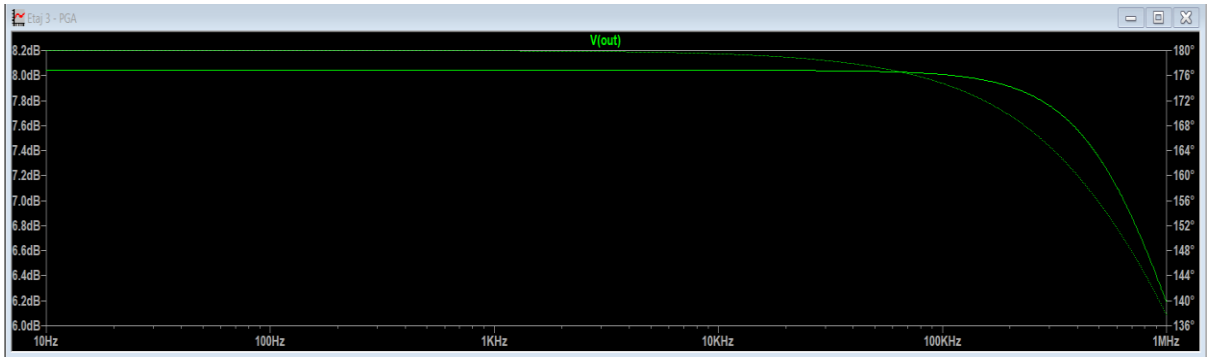


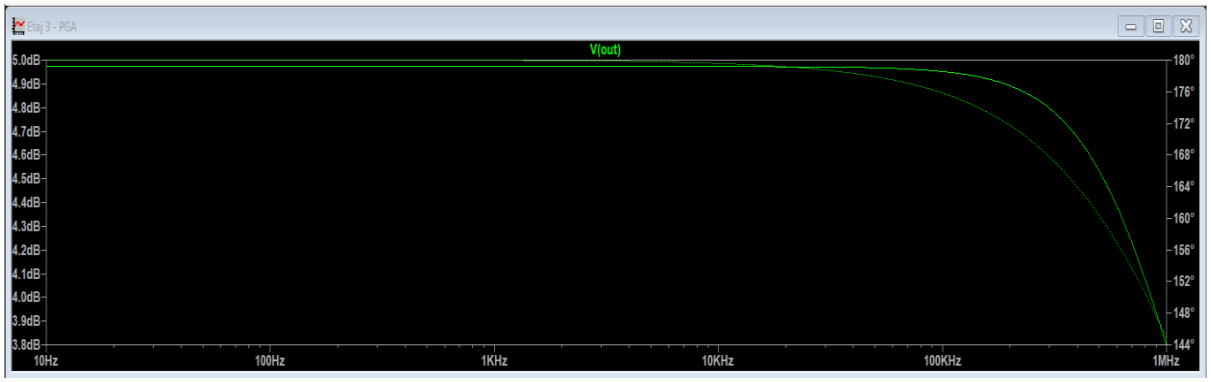
Simulare: DCOP

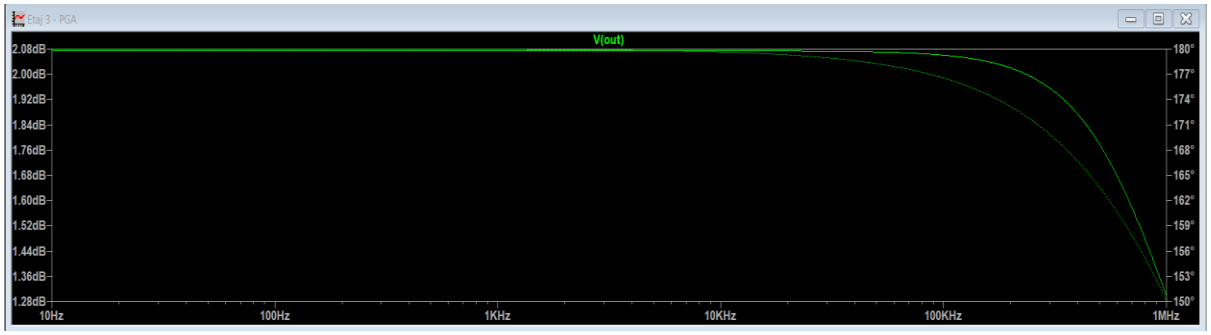
C:\Users\Lenovo\Desktop\.Proiect - SIMULARI\Etaj 3 - PGA.net

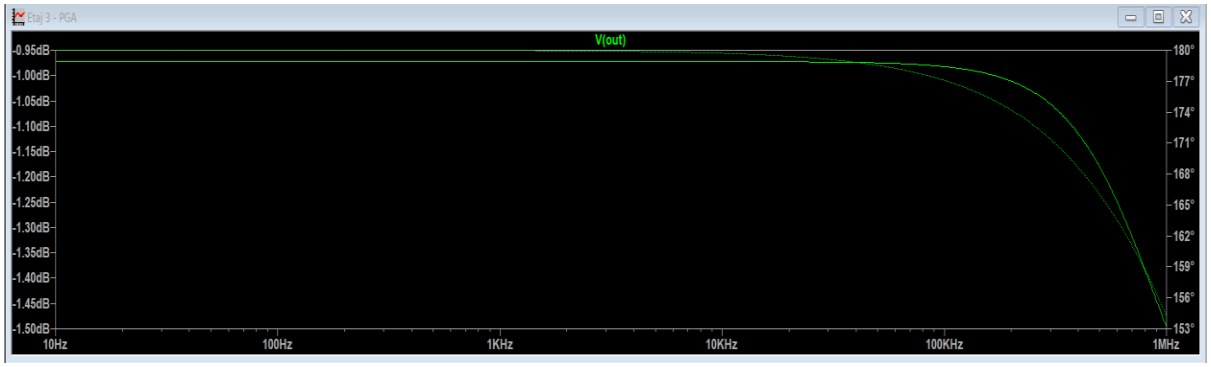
```
        --- Operating Point ---

V(n001):        3.99944         voltage
V(out):         14.0959         voltage
V(n002):        14.0959         voltage
V(n004):        14.0959         voltage
V(s3):          0               voltage
V(s2):          0               voltage
V(vdd):         15              voltage
V(in):          0               voltage
V(n003):        14.0959         voltage
V(n005):        4.04912         voltage
V(s1):          0               voltage
V(s4):          15              voltage
I(R3):          2.42023e-11     device_current
I(R4):          0.00121191      device_current
I(V5):          0               device_current
I(V7):          0               device_current
I(V2):          -0.00100412     device_current
I(R1):          2.42023e-11     device_current
Id(M3):         2.42023e-11     device_current
Ig(M3):         0               device_current
Is(M3):         -6.08699e-12    device_current
Ib(M3):         -1.81153e-11    device_current
Id(M2):         2.42023e-11     device_current
Ig(M2):         0               device_current
Is(M2):         -6.08699e-12    device_current
Ib(M2):         -1.81153e-11    device_current
I(R2):          2.42023e-11     device_current
Id(M1):         2.42023e-11     device_current
Ig(M1):         0               device_current
Is(M1):         -6.08699e-12    device_current
Ib(M1):         -1.81153e-11    device_current
Id(M4):         0.00121191      device_current
Ig(M4):         0               device_current
Is(M4):         -0.00121191     device_current
Ib(M4):         -8.06856e-12    device_current
I(RG):          0.00121195      device_current
I(V4):          0               device_current
I(V6):          0               device_current
I(Vin):         0.00121195      device_current
Ix(u1:1):       -8.33329e-08    subckt_current
Ix(u1:2):       -3.86358e-08    subckt_current
Ix(u1:3):       0.00100412      subckt_current
Ix(u1:4):       -0.00100413     subckt_current
Ix(u1:5):       -0.00121191     subckt_current
```

Simulare: Câştig [-1, 2, 5, 8]









Simulare: Liniariate

SPICE Output Log: C:\Users\Lenovo\Desktop\.Proiect - SIMULARI\Etaj 3 - PGA.log

```
Files loaded:
C:\Users\Lenovo\Desktop\.Proiect - SIMULARI\Etaj 3 - PGA.net
C:\Users\Lenovo\AppData\Local\LTspice\lib\cmp\standard.mos
C:\Users\Lenovo\Desktop\.Proiect - SIMULARI\tsmc180nmcmos.lib
C:\Users\Lenovo\AppData\Local\LTspice\lib\sub\OP482.lib

Fourier components of V(OUT)
N-Period=1
DC component:7.57618

Harmonic            Frequency           Fourier            Normalized          Phas
 Number               [Hz]              Component            Component          [degr
    1               1.000e+03           1.788e-01           1.000e+00           -89
    2               2.000e+03           1.986e-07           1.111e-06          -178
    3               3.000e+03           1.267e-08           7.087e-08            58
    4               4.000e+03           2.413e-08           1.349e-07           -14
    5               5.000e+03           5.081e-08           2.842e-07           -67
    6               6.000e+03           2.909e-08           1.627e-07           166
    7               7.000e+03           2.737e-08           1.530e-07           133
    8               8.000e+03           4.648e-08           2.599e-07            13
    9               9.000e+03           3.559e-08           1.990e-07           -56
   10               1.000e+04           1.012e-08           5.662e-08          -156
Partial Harmonic Distortion: 0.000122%
Total Harmonic Distortion:   0.000659%

Fourier components of V(OUT)
N-Period=1
DC component:7.57618

Harmonic            Frequency           Fourier            Normalized          Phas
 Number               [Hz]              Component            Component          [degr
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    3               3.000e+03           1.267e-08           7.087e-08            58
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    5               5.000e+03           5.081e-08           2.842e-07           -67
    6               6.000e+03           2.909e-08           1.627e-07           166
    7               7.000e+03           2.737e-08           1.530e-07           133
    8               8.000e+03           4.648e-08           2.599e-07            13
    9               9.000e+03           3.559e-08           1.990e-07           -56
   10               1.000e+04           1.012e-08           5.662e-08          -156
Partial Harmonic Distortion: 0.000122%
Total Harmonic Distortion:   0.000659%
```
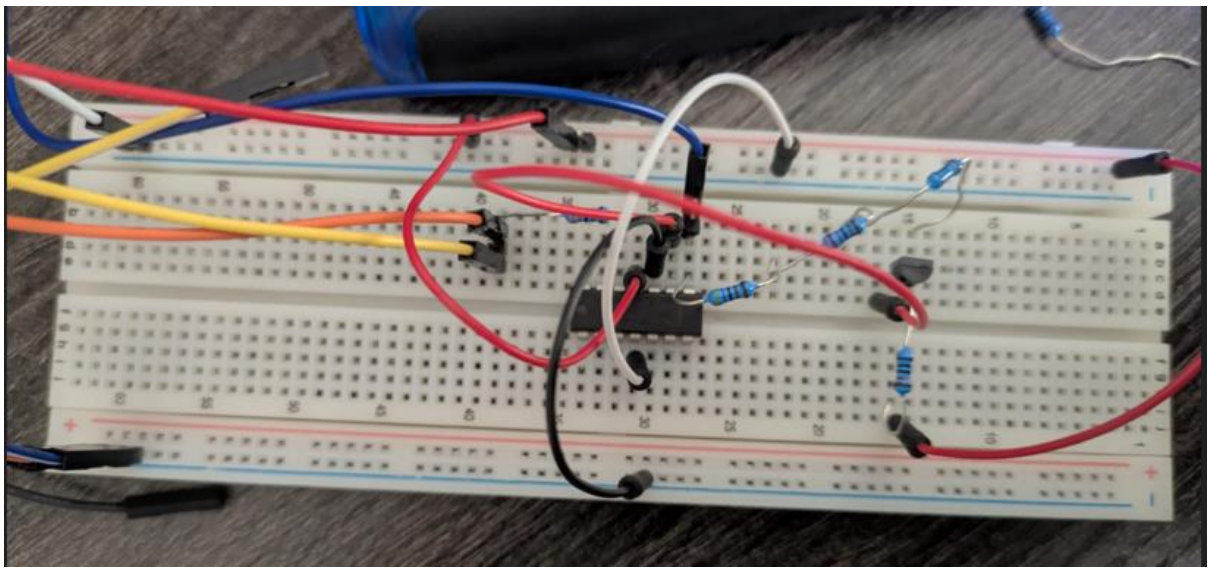
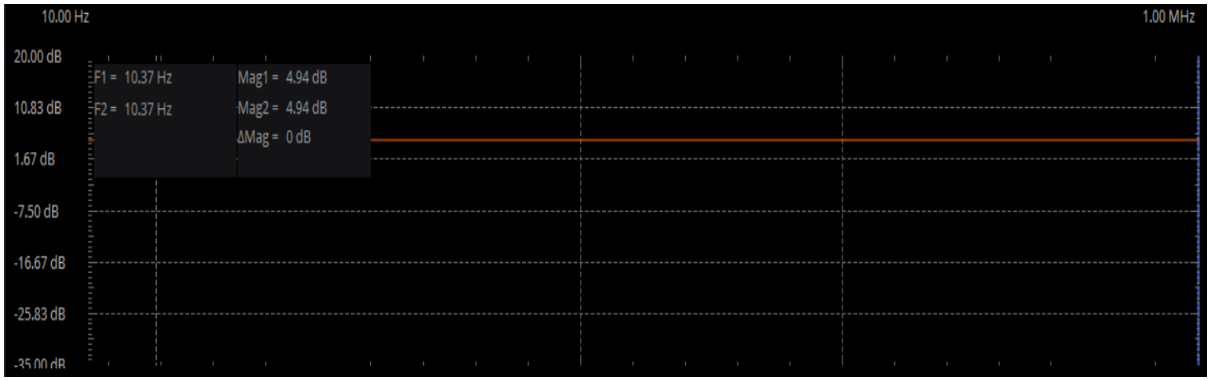
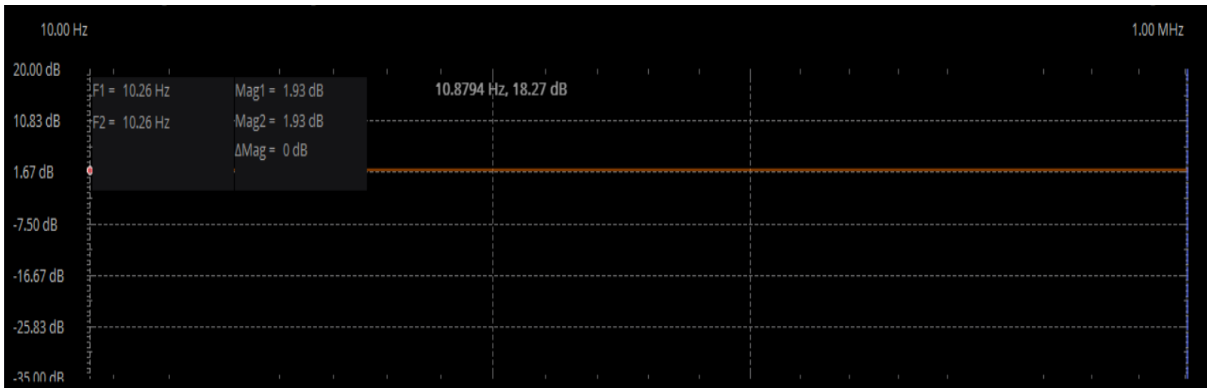
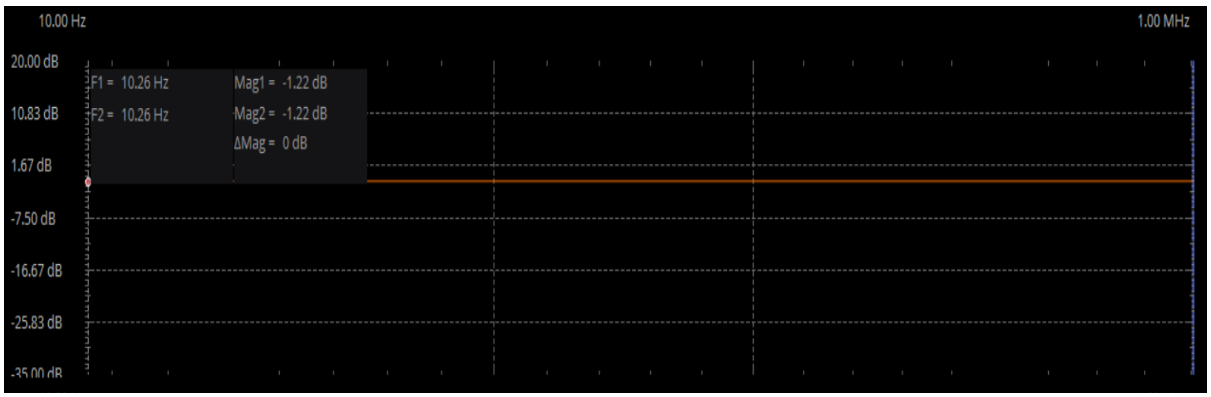

Simulare: Circuit Real



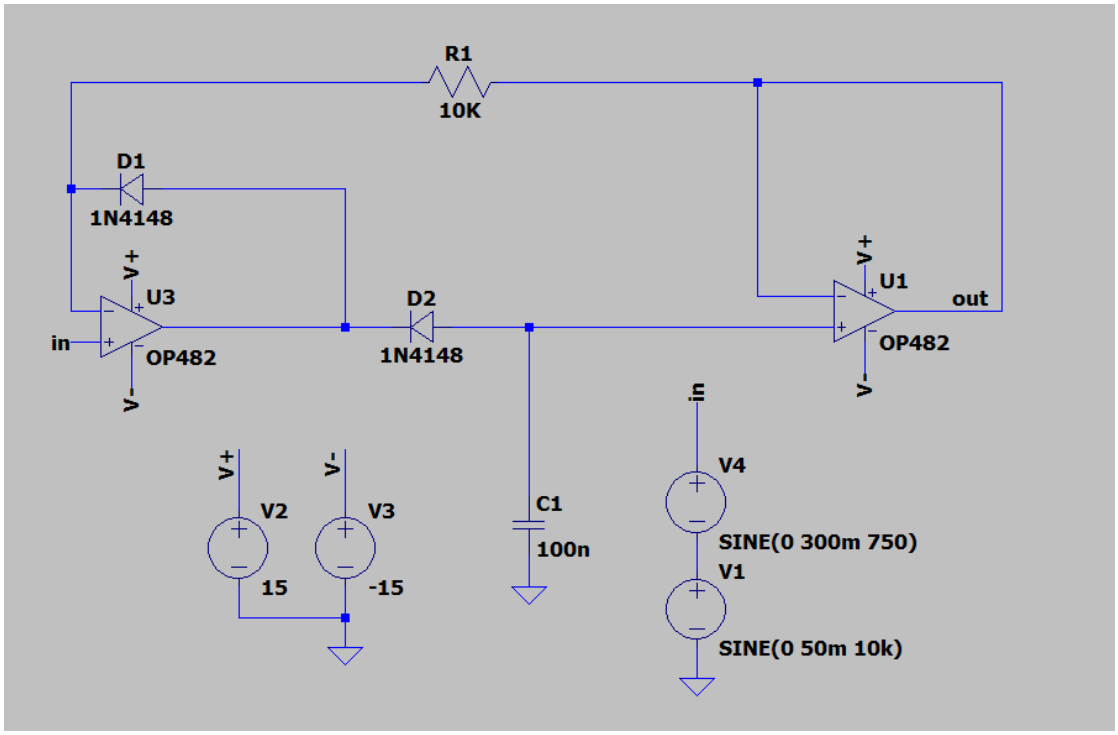
Simulare: Câștig



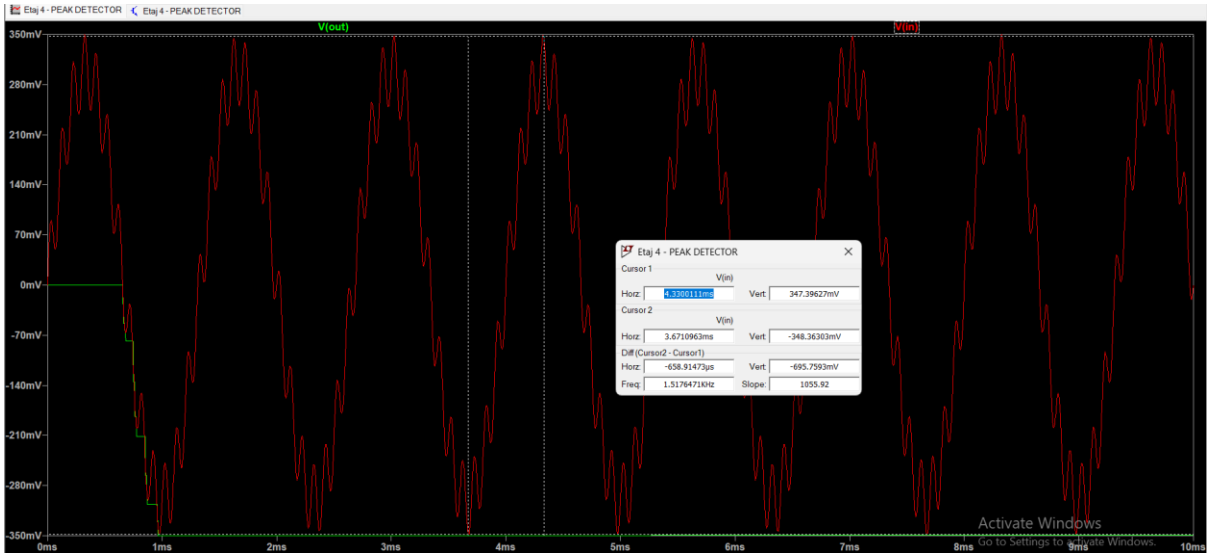


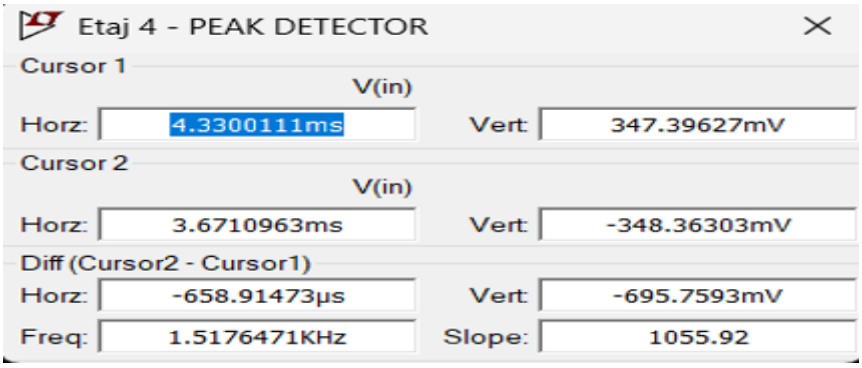


- Etaj 4:

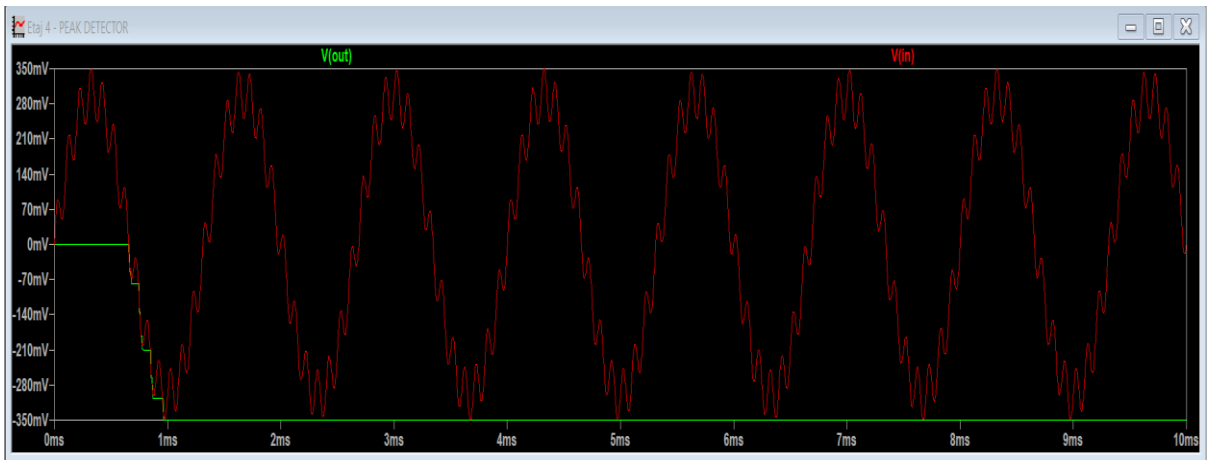


Simulare: Câştig

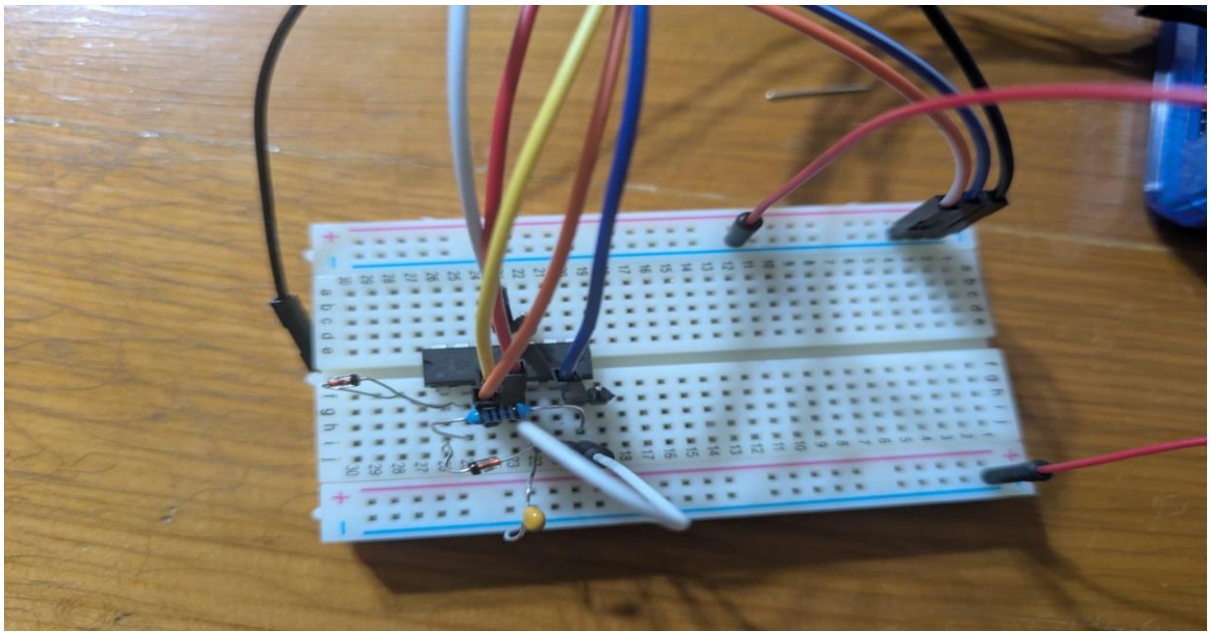


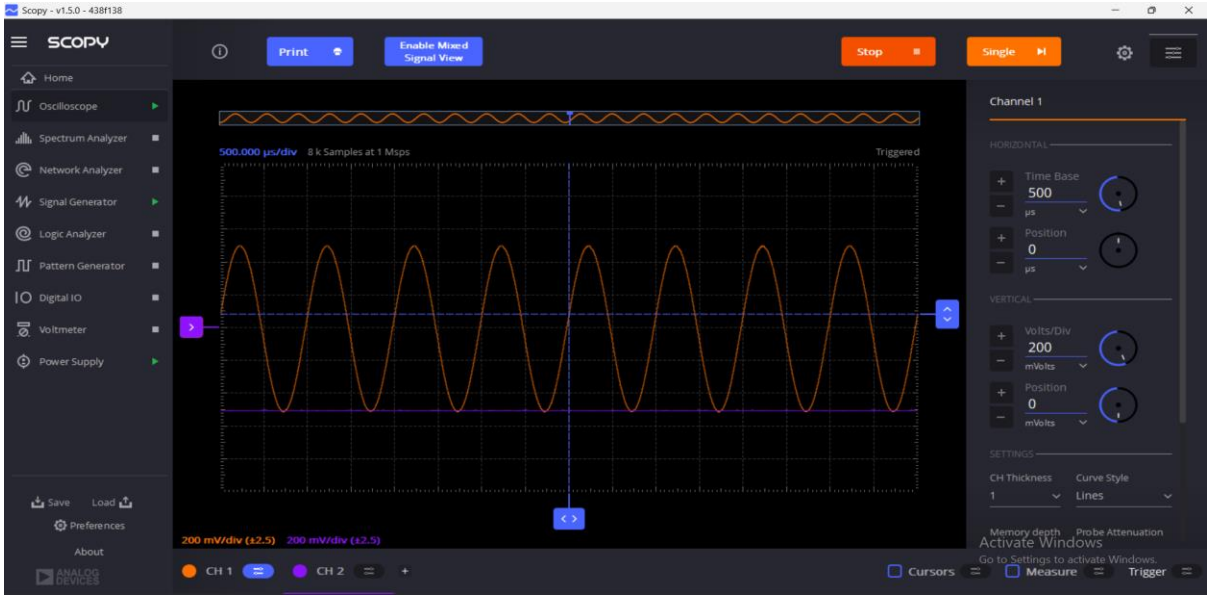


Simulare: Funcție de circuit

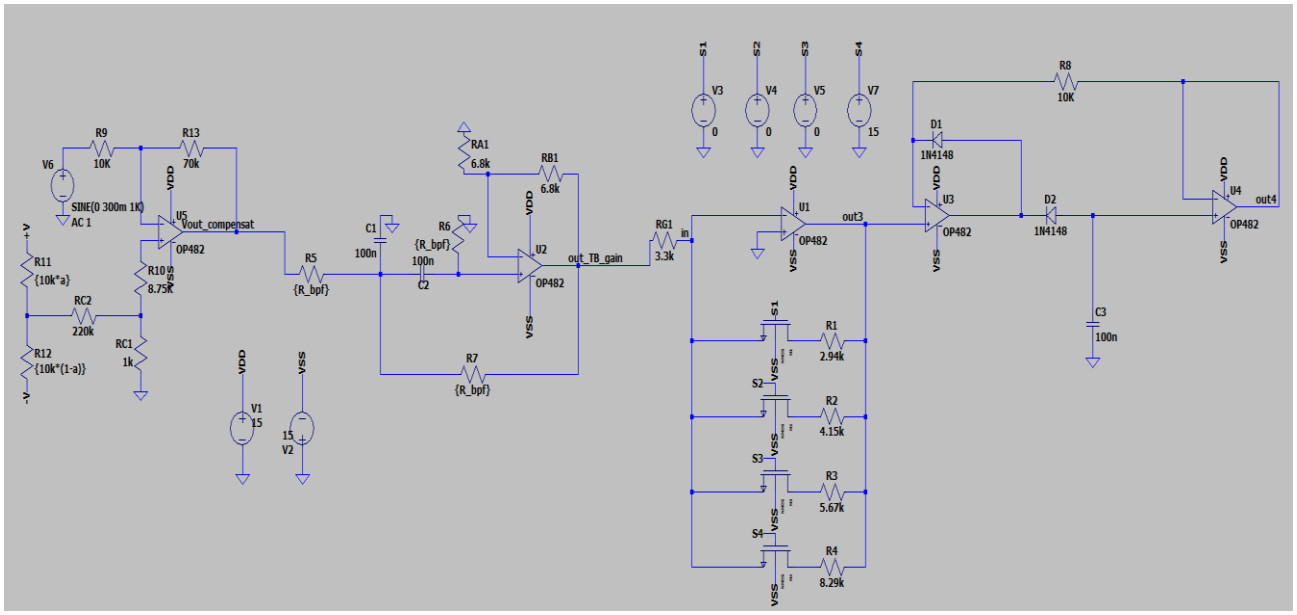


Simulare: Circuit Real

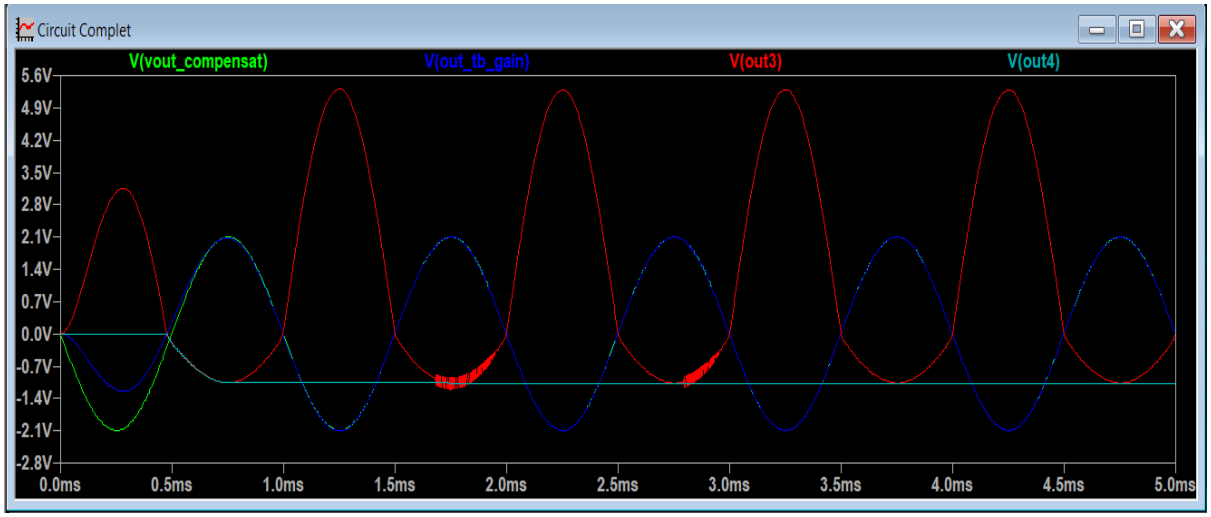




- Circuit Complet:



Simulare:



Verificarea şi caracterizarea interfeţei analogice

- Funcționare logică:

Semnalul de intrare va trece prin fiecare etaj, acesta va fi supus la modificari.

-Etaj 1: Vin minim = 199mV -> Vout minim AO = 1,393V

Vin maxim = 561mV -> Vout maxim AO = 3,923V

-Etaj 2: Filtrarea

-Etaj 3:

*pentru câştig Vout etaj 1 minim

Vout minim PGA => 1,393 * 2,51 = 3,499V

Vout maxim PGA => 3,923 * 2,51 = 9,854V

*pentru câştig Vout etaj 1 maxim

Vout minim PGA => 1,393 * 0,891 = 1,241V

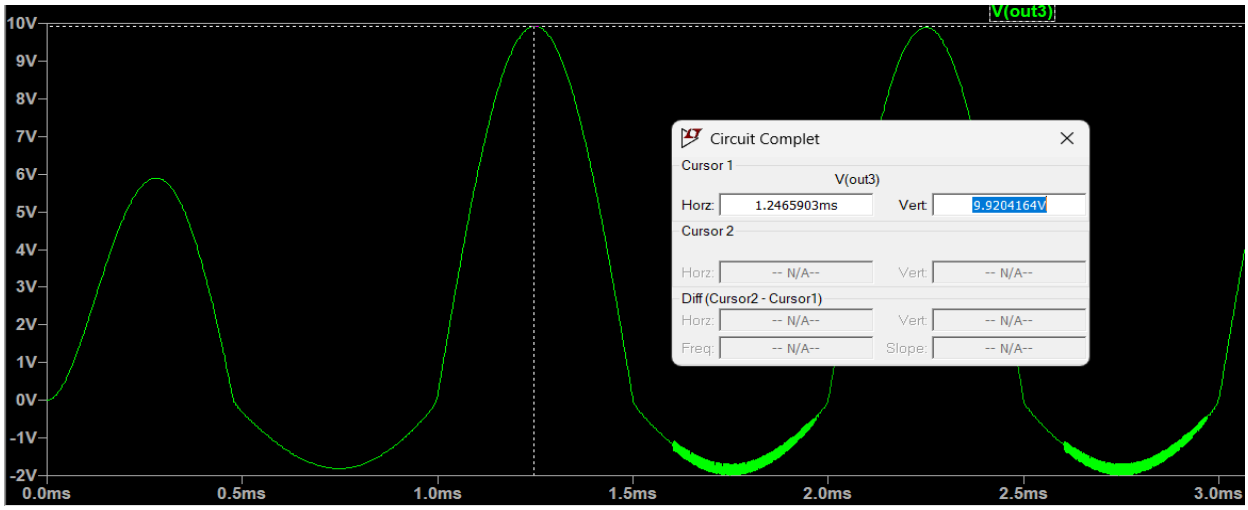
Vout maxim PGA => 3,923 * 0,891 = 3,495V

+alte ieşiri in funcție de valorile alese.

-Etaj 4: Redresarea

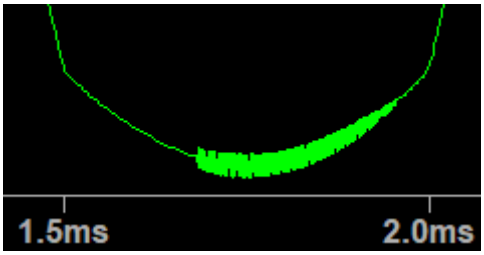
- Verificarea punctului de saturație

Interfața funcționează în regim liniar deoarece amplitudinea maximă la ieșire (aprox. 9.84V în cazul câștigului maxim) se află sub pragul de saturare al amplificatoarelor operaționale, asigurând o distorsiune minimă



- Analiza stabilității

S-a observat o degradare a stabilității buclei de reacție odată cu creșterea amplitudinii semnalului; mai exact, peste 300mV, capacitățile parazite ale switch-urilor NMOS provoacă instabilitate dinamică. Acest comportament afectează direct precizia conversiei AC-DC, generând un offset nedorit la ieșirea finală. In timp ieșirea se va repera.



Comparare rezultate simulari - măsurători

ETAJ 1

VOUT compensat	SIMULĂRI	MĂSURĂTORI
199 mV	1,393 V	1,389 V
561 mV	3,926 V	3,921 V

ETAJ 3

CÂȘTIG - 561mV	SIMULĂRI	MĂSURĂTORI
-1 dB	3,529 V	3,47 V
2 dB	4,967 V	4,89 V
5 dB	6,797 V	6,71 V
8 dB	9,881 V	9,81 V

CÂȘTIG - 199mV	SIMULĂRI	MĂSURĂTORI
-1 dB	1,256 V	1,249 V
2 dB	1,776 V	1,71 V
5 dB	2,411 V	2,36 V
8 dB	3,519 V	3,45 V

Concluzii

Proiectul a vizat realizarea unui sistem electronic complex structurat în patru etaje funcționale, utilizând amplificatoare operaționale de precizie tip **OP482**. În urma etapelor de proiectare, simulare și testare pe circuit real.

Primul etaj a asigurat cu succes amplificarea cu un câștig liniar de 7, integrând un mecanism eficient de compensare a tensiunii de offset prin reglaj potențiometric.

Etajul al doilea a demonstrat o selectivitate ridicată, funcționând ca un filtru trece-bandă cu o frecvență centrală de 1kHz și un factor de calitate Q = 0.707.

Etajul al treilea consta in implementarea amplificatorului cu câștig variabil a permis ajustarea amplitudinii semnalului în patru pași (între -1dB si 8dB), cu o rezoluție de 3dB, oferind versatilitate sistemului în funcție de amplitudinea sursei de semnal.

Etajul al patrulea a asigurat cu succes funcția de detector de vârf negativ, utilizând o configurație de redresor de precizie cu câștig liniar de 1 pentru a extrage valoarea minimă a semnalului.

În concluzie, obiectivele proiectului au fost îndeplinite, realizându-se un sistem capabil de condiționare, filtrare, amplificare selectabilă și redresare de precizie a semnalelor analogice.